Does Argument Structure Help Essay Scoring?
RQ1 and RQ2 on PERSUADE 2.0 · Week 10 progress

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Group 04

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1. Scope and what we answered

Scope after the feedback

- **Core:** Step 3, essay scoring, RQ1 and RQ2
- **Fixed design:** 8 independent prompts, leave-one-prompt-out, Gradient Boosting, 3 seeds
- **Side result:** Step 2 (segment effectiveness). Fairness (RQ3) only after RQ1 and RQ2
- **Claim we make:** gold structure adds a little (+0.02 QWK); predicted roles add nothing beyond segment boundaries

Data in this study

7,869	77,639	1.9%
essays · 8 prompts	labelled segments	segments Ineffective

- Essay: holistic score 1–6. Segment: role + effectiveness
- Seven source-based prompts excluded: their scores depend on the source text
- Official test set untouched until the end

2. Data workflow

From raw file to model input

- Raw CSV: one row per segment, full essay text repeated on every row (about 600 MB)
- **Essay table:** drop duplicates → 15,594 essays, one score each
- **Segment table:** 173,266 segments with role, effectiveness, position
- **Filter:** task = Independent → 7,869 essays, 8 prompts
- **Features:** 43 text features per essay (regular expressions) + 28 structure features from roles

Why each choice

Choice	Reason
One row per essay	the essay is the unit that gets a score
Independent prompts only	do not mix with source-based scoring
Regex features	each feature is countable and can be explained
Split by prompt	the question is a new prompt, not a new essay
Role features only in M2, M3	so M2/M3 differ from M1 in one block

Ineffective is 1.9% of segments in these 8 prompts (4.5% in all 15).

3. One essay, step by step

Essay F3CB6601CF24 · Seeking multiple opinions

- **Segments (gold):** Lead 203 · Position 20 · Claim 18 · Evidence 120 · Claim 21 · Evidence 112 · Claim 17 · Evidence 110 · Concluding 117 words
- **M1 features:** 741 words and 3,928 characters (8-prompt mean: 464 and 2,591)
- **Structure:** 3 Claims, 3 Evidence → evidence_per_claim = 3 / 3 = 1.0
- **Length shortcut:** mean essay length rises from 272 words (score 1) to 854 words (score 6)

What each model predicts (true score 5)

Model	Input	Prediction
M0a	word count	5.07
M1	43 features	4.62
M2	+ gold roles	4.33
M3	+ predicted roles	4.10

A case where structure moves the score away from the truth. One case shows the mechanism, it is not evidence.

4. Models and formulas

Formulas

- **QWK** = $1 - \Sigma\, w \cdot O / \Sigma\, w \cdot E$, $w = (i - j)^2 / 25$: big misses cost more
- **Macro-F1** = mean F1 over classes: Ineffective is rare
- **evidence_per_claim** = n_Evidence / max(n_Claim, 1)
- **Flesch** = $206.835 - 1.015 \cdot (\text{words} / \text{sentences}) - 84.6 \cdot (\text{syllables} / \text{words})$
- **Class weights**: $\hat{y} = \text{argmax}\ w_k \cdot P(k)$, w tuned on training prompts only

Models and length controls

	Input	Role
M0a	word count	old baseline
M0b	7 length features	length-only ceiling
M1	43 features	baseline
M1 + B	M1 + boundaries	what M3 still knows
M2	M1 + gold roles	oracle
M3	**M1 + predicted roles**	**realistic**

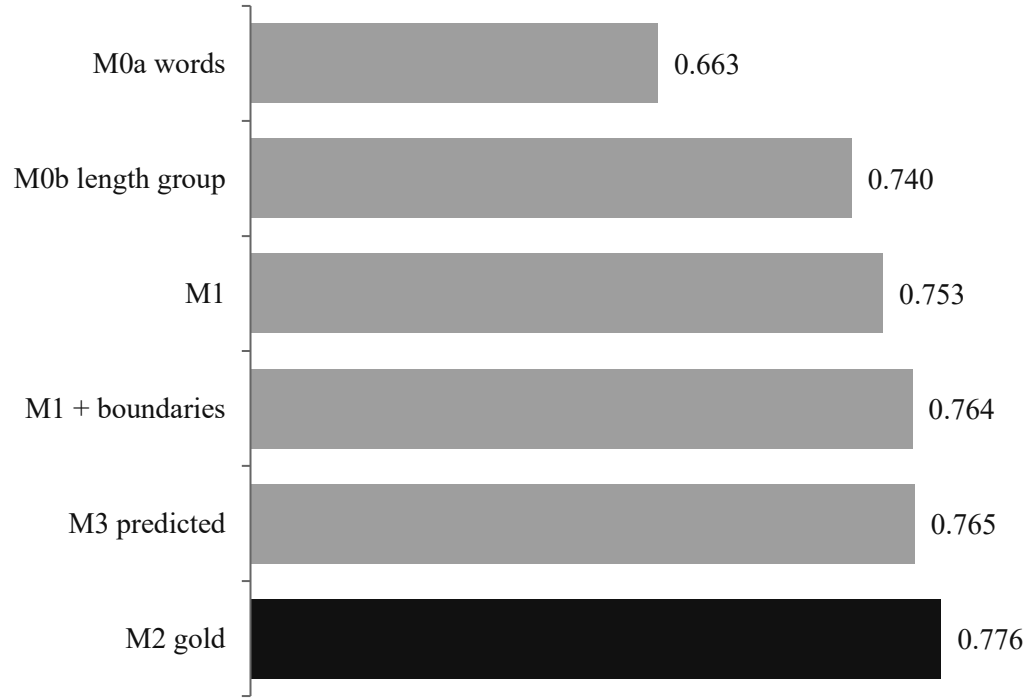
Same algorithm, same prompt-based folds: only the inputs differ.

5. Length explains most of M1

From word count to structure

- **M1 − M0a = 0.090:** 0.077 (86%) is already in the length group
- **M1 − M0b = +0.013:** language beyond length is small
- **M2 − M1 = +0.023:** gold structure, positive in 8 of 8 prompts
- **M3 − M1 = +0.012:** predicted roles, 7 of 8 prompts, p = 0.07
- **M1 + B − M1 = +0.012:** boundaries alone give as much as M3

Mean QWK per held-out prompt



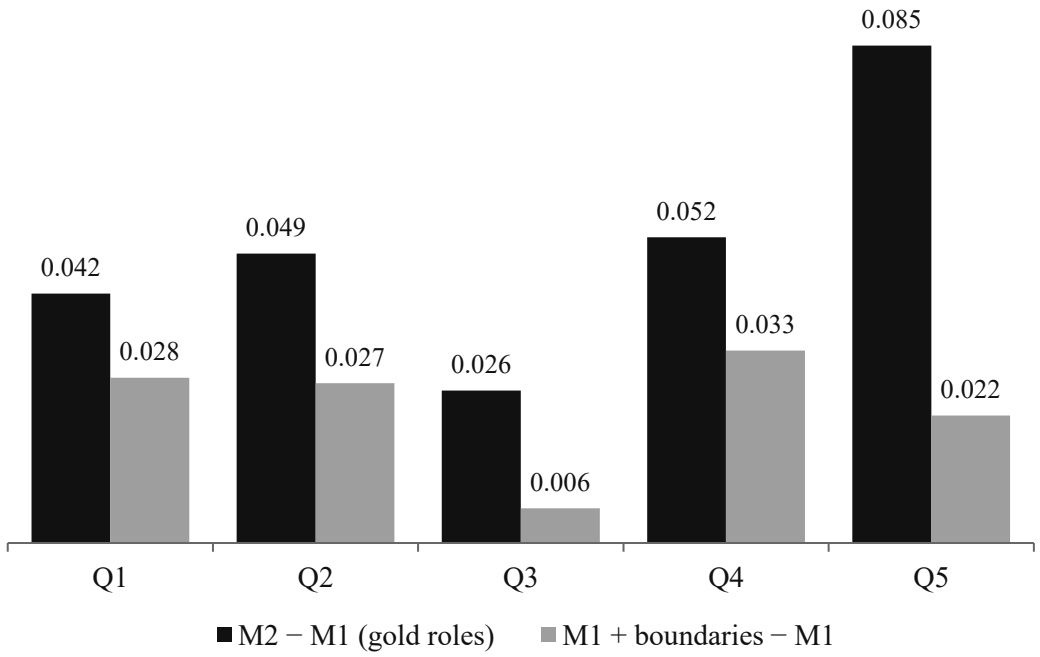
Gradient Boosting, 3 seeds, 8 held-out prompts.

6. Structure inside length groups

Five groups of equal size by word count

- About 1,570 essays per group (Q1 shortest, Q5 longest)
- **Gold structure helps in all five groups:** +0.026 to +0.085 QWK
- In the shorter groups, boundaries alone give most of the gain
- **In the longest group** roles add +0.063 beyond boundaries (95% CI 0.036 to 0.086)
- Length control is partial: score still varies inside each group

ΔQWK against M1 in each group



Bootstrap over essays, 500 draws; GB predictions from leave-one-prompt-out.

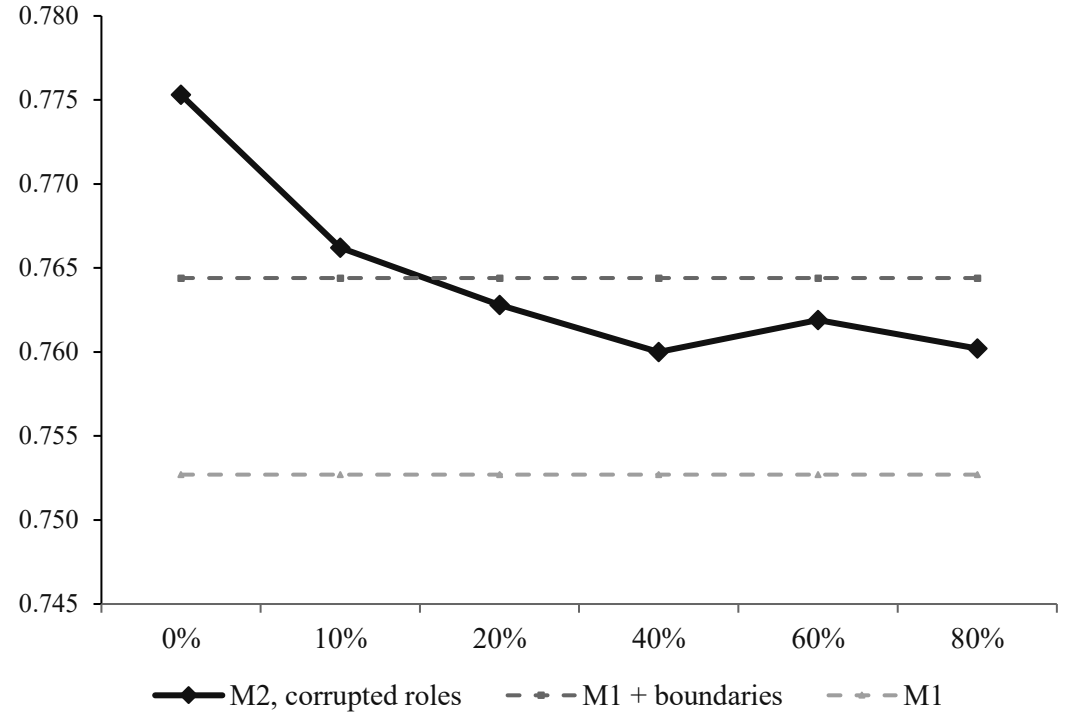
7. How good must the role tagger be?

Corrupt gold role labels, keep boundaries

- Change 10, 20, 40, 60, 80% of role labels at random; rerun M2 on the same folds; 3 draws per level
- **At 20%** M2 falls below M1 + boundaries
- **From 40%** the curve is flat, still above M1 because boundaries are intact
- **Our tagger** (macro-F1 0.675) is already in this zone

RQ2: with the current tagger, predicted roles do not beat "count the paragraphs".

M2 mean QWK by % labels corrupted



Random replacement among the 7 roles is not how a real tagger errs (Claim and Evidence get confused).

8. Step 2: segment effectiveness

Better, but not a teacher flag

- **Baseline:** macro-F1 0.433, F1 of Ineffective 0.07
- **Improved:** role, length, position, essay context, text signal, class weights tuned on training prompts → 0.510 pooled, 0.477 per prompt
- **Embeddings and context similarity add nothing** (0.473 vs 0.477 per prompt)
- **Ranking:** flag the top 5% of segments → 9.3% correct (4.7× chance), catches 23% of Ineffective

Step 2, leave-one-prompt-out

Model	Macro-F1 pooled	per prompt	F1 Ineff.	PR-AUC
Majority class	0.291	0.293	0.000	n/a
Week-8 baseline	0.433	0.413	0.070	n/a
S3 tuned	0.509	0.477	0.120	0.072
S6 tuned	0.513	0.473	0.123	0.074

Above 0.5 only pooled. Precision of Ineffective is about 0.10: nine flags in ten are wrong. Chance for PR-AUC is 0.022.

9. Related work, 2024–2026

What recent papers say

- **Yang et al. (2024), AAAI:** 9 scorers, accuracy and fairness; feature-based models generalise well
- **Wang et al. (2026), ArguLens:** gold discourse features +0.055 QWK, an oracle diagnostic
- **Gui (2026), AIME-Con:** effectiveness signal tracks length and elaboration
- **Ormerod (2025):** argument annotations help encoder scorers
- **Keramati & Warschauer (2026):** sequential LoRA fine-tuning on discourse elements
- **Chen, Wu & Zhang (2026):** conditional disparity ratio and difference at the same true score; our plan for RQ3

How we use them

- Hand-made features + trees, prompt-based split: the setting Yang et al. recommend
- M2 = oracle, M3 = intermediate: the ArguLens distinction
- Step 2 result matches Gui: little beyond role, length, position

Not compared: our QWK 0.753–0.776 and ArguLens 0.813 use different data, features and splits.

10. Working with Group 6

Their reply and our decision

- **They:** Step 2 with DeBERTa, random 80/20 split on 15 prompts, macro-F1 0.577
- **We:** Step 3, leave-one-prompt-out on 8 prompts
- **Copy the procedure, not the numbers:** different splits
- **Their validation file would leak:** their model saw our held-out prompts
- **No GPU on our side:** ask for 1–2 DeBERTa folds

What we exchange

We send	They send
List of our 8 prompts	Leave-one-prompt-out probabilities (RF / LR) on those 8 prompts
Predicted roles (macro-F1 0.675)	DeBERTa on 1–2 folds
Essay-level features	Their error-propagation result with predicted roles

Draft minutes in the report; Group 6 still has to confirm.

11. Limits and next steps

Limits

- Only 8 prompts; more seeds do not add independent prompts
- Segment boundaries are gold: M3 is not an end-to-end tool
- Corruption test: one GB seed per draw, uniform random errors
- Step 2 gain is mostly a changed decision threshold
- Six cases not yet checked against the rubric

Next steps

- Re-read the six cases with the rubric; add an essay with Claim and Evidence in few paragraphs
- Replace gold boundaries with predicted ones
- Confirm which kappa the human ceiling 0.75 is (ETS treats QWK around 0.70 as acceptable, via Chen et al.)
- DeBERTa fold from Group 6; then RQ3

Claim: on 8 prompts, gold structure adds about +0.02 QWK; predicted roles add nothing beyond boundaries yet.